

Diffraction of light : Bending of light around the corners of an obstacle whose size is comparable to the wavelength of light is called diffraction. (The spreading of light into the geometrical shadows)

Diffraction of sound waves is more ^{common} evident in our daily life than that of light waves.

⇒ For obtaining a well defined diffraction pattern, the size of the obstacle/aperture should be of the same order as the wavelength of the wave. Wavelength of light ($\approx 10^{-7} \text{ m}$) is very small as compared to the size of the obstacles/apertures (corners of buildings, wide openings etc) encountered in daily life.

Interference

Due to superposition of waves from two different wavefronts originating from two coherent sources.

All fringes (bright & dark) are of equal widths.

All bright fringes are of equal intensity.

Diffraction

1) Due to superposition of secondary waves originating from different parts of the same wavefront.

2) The width of the central bright fringe is much more than that of any secondary maxima.

3) Intensity $\propto$ as we the order of diffraction is.

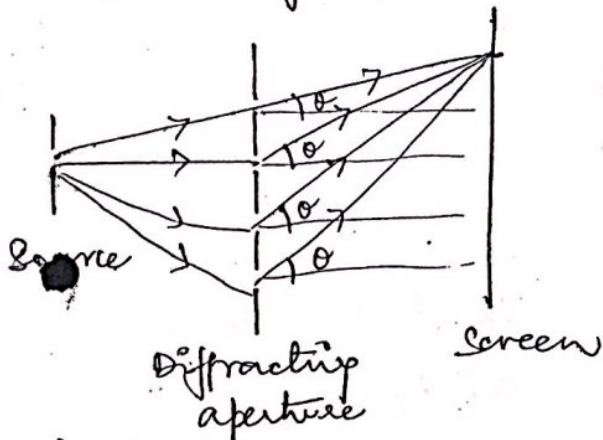
and contrast betn
& bright fringes is
(dark regions are
completely dark.)

4) ~~poor~~ Poor contrast $\rightarrow$ dark
regions are not
completely dark.

Types of Diffraction of Light waves.

Fresnel Diffraction

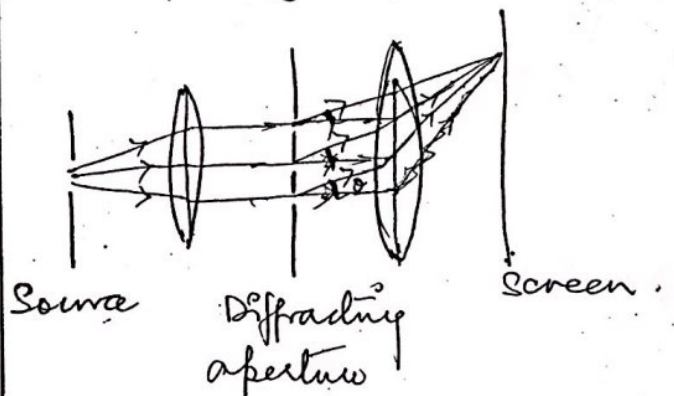
- 1) The source of light & the screen are at a finite distance from the diffracting aperture.



- 2) Incident wavefronts are divergent (spherical / cylindrical)
- 3) No mirror or lens is used for observation.
- 1) The centre of diffraction pattern may be bright or dark.

Fraunhofer Diffraction

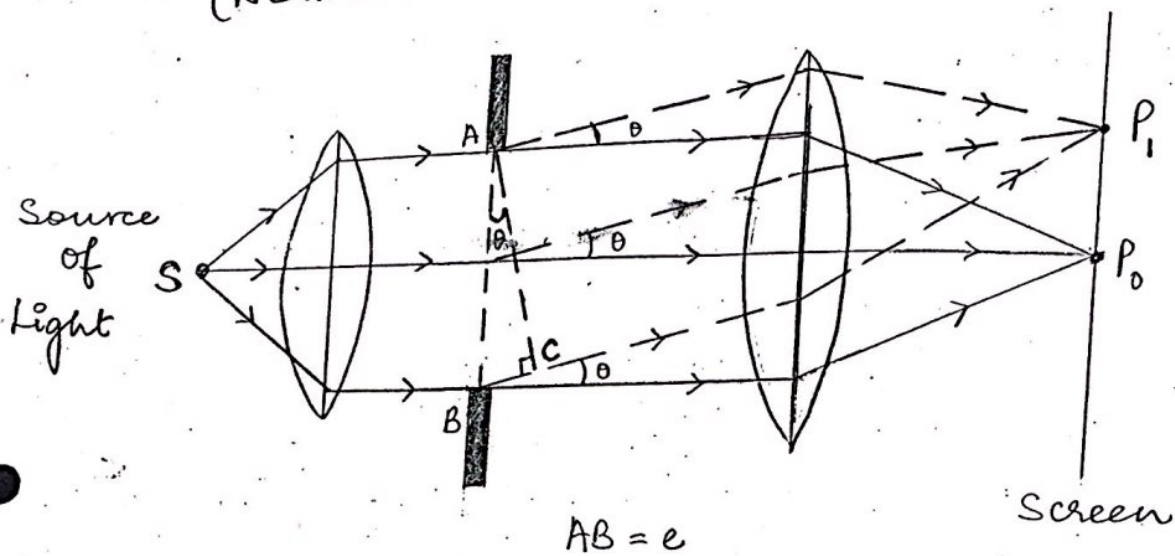
- 1) The source of light & the screen are at ∞ distance from the diffracting aperture.



- 2) Incident wavefronts are plane
- 3) Diffracted light is collected by a lens in the telescope
- 4) The centre of diffraction pattern is always bright for all paths parallel to the ~~wave~~ incident light.

DIFFRACTION OF LIGHT.

Fraunhofer diffraction at a single slit (Normal incidence)



Consider a narrow slit AB of width ' e ' illuminated by monochromatic light of wavelength ' λ ' falling normally on it. Let the diffracted light be focussed by means of a convex lens on a screen placed in the focal plane of the lens. According to the Huygen's principle, every point of the wavefront from the slit is a source of secondary wavelets which spread out in all directions. The secondary wavelets in the direction OP_0 are focussed at point P_0 on the screen and P_0 is the central bright fringe. The secondary wavelets diffracted at angle ' θ ' are brought to focus at point P_1 .

Path difference between secondary wavelets from A & B at angle ' θ ',

$$\Delta = BC = AB \sin \theta = e \sin \theta$$

$\therefore$ Corresponding phase difference,

$$\phi = \frac{2\pi}{\lambda} \cdot \Delta$$

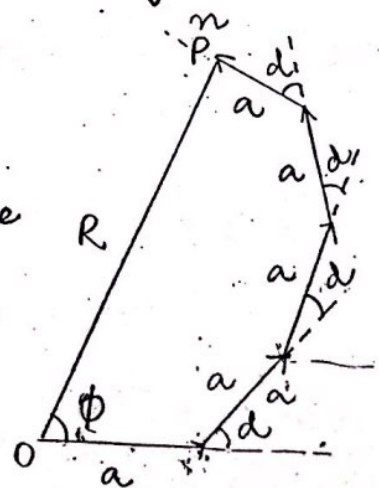
$$\phi = \frac{2\pi}{\lambda} e \sin \theta.$$

Let the slit be divided into ' n ' equal parts and the amplitude of the wave from each part be ' a '.

$\therefore$ Phase diff. between any two consecutive waves from these parts $\Rightarrow \frac{1}{n}$ (Total Phase)

$$= \frac{1}{n} \left(\frac{2\pi}{\lambda} e \sin \theta \right) = d \text{ (say)}$$

n vibrations of same period, same amplitude ' a ' and same phase difference ' d ' between successive vibrations.



The resultant amplitude of these vibrations by phasor notation (vector addition) is given by,

$$R = \frac{a \sin\left(\frac{nd}{2}\right)}{\sin\left(\frac{d}{2}\right)}$$

$$= \frac{a \sin(\pi e \sin \theta / \lambda)}{\sin(\pi e \sin \theta / n \lambda)}$$

$$= a \frac{\sin \alpha}{\sin(\alpha/n)} \quad \left(\text{let } \alpha = \frac{\pi e \sin \theta}{\lambda} \right)$$

when $n \rightarrow \infty$, $\frac{\alpha}{n}$ becomes very small,

$$\therefore \sin \frac{\alpha}{n} \approx \frac{\alpha}{n}$$

$$\therefore R = na \frac{\sin \alpha}{\alpha}$$

$$na = A \text{ (total amplitude)}$$

• [when $n \rightarrow \infty$, $a \rightarrow 0$, $na = A$ remains finite]

$$\therefore \text{Resultant amplitude, } R = A \frac{\sin \alpha}{\alpha}$$

$$\text{Intensity, } I = R^2 = A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2$$

$$I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2, \quad I_0 = A^2$$

Intensity distribution in single slit pattern:
positions of Maxima & Minima:

$$R = A \left(\frac{\sin \alpha}{\alpha} \right)$$

$$= \frac{A}{\alpha} \left[\alpha - \frac{\alpha^3}{3!} + \frac{\alpha^5}{5!} - \frac{\alpha^7}{7!} + \dots \right]$$

$$= A \left[1 - \frac{\alpha^2}{3!} + \frac{\alpha^4}{5!} - \frac{\alpha^6}{7!} + \dots \right]$$

For $R_{\max}$, all negative terms must vanish

$$\therefore \alpha = 0,$$

$$\therefore R_{\max} = A \quad \text{for } \underline{\alpha = 0}$$

$$\alpha = \frac{\pi e \sin \theta}{\lambda} = 0 \quad \Rightarrow \quad \underline{\sin \theta = 0}$$

$\therefore$ For $\theta = 0$, for secondary wavelets that travel normally from the slit & reach point P, we have the maxima with intensity proportional to A^2 . & is known as principle maxima.

* Minimum intensity positions,

when $\sin \alpha = 0$, $R = 0$ but $\alpha \neq 0$

$$\therefore \alpha = \pm \pi, \pm 2\pi, \dots, \pm n\pi$$

$$\therefore \left(\frac{\pi e \sin \theta}{\lambda} \right) = \pm n\pi$$

$$e \sin \theta_n = \pm n\lambda, \quad n=1, 2, \dots$$

Points of minimum intensity are obtained on either side of central maxima.

Note: Value of $n=0$ is not admissible, $\because$ for this $\theta=0$ which corresponds to the principle maxima.

Secondary maxima:

$$I = A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2$$

For maxima, $\frac{dI}{d\alpha} = 0$
or minima

$$\Rightarrow A^2 \left(\frac{2 \sin \alpha}{\alpha} \right) \left[\frac{\alpha \cos \alpha - \sin \alpha}{\alpha^2} \right] = 0$$

$\Rightarrow$ either $\sin \alpha = 0$ or $(\alpha \cos \alpha - \sin \alpha) = 0$

$\sin \alpha = 0$ corresponds to positions of minimum intensity.

$\therefore$ For Secondary maxima, $\alpha \cos \alpha - \sin \alpha = 0$

$$\Rightarrow \alpha = \tan \alpha.$$

Graphically, the values of α can be determined by plotting the curves $y = \alpha$ & $y = \tan \alpha$ on the same graph.

α values of α are the pts of intersection of two curves given by,

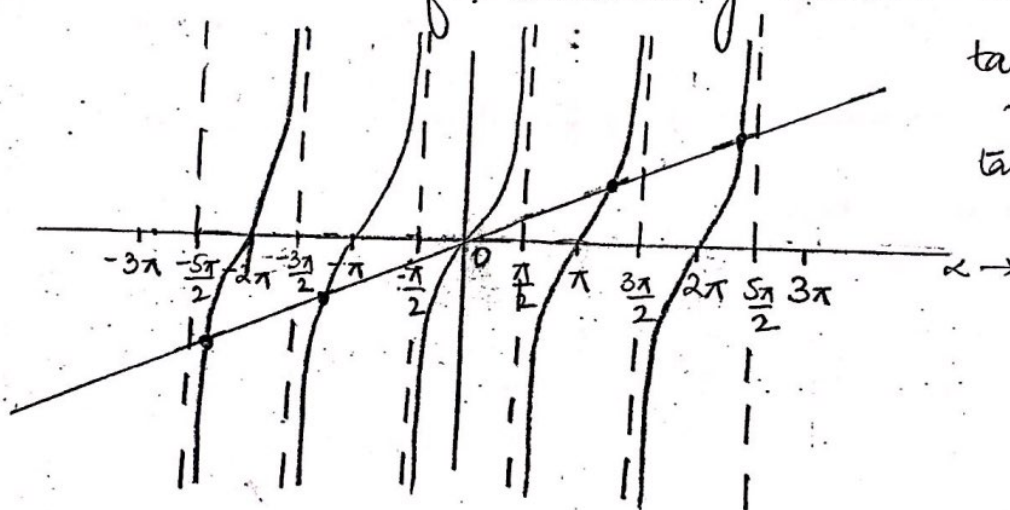
$$\alpha = 0, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2} \dots$$

Note: $\alpha = 0$ corresponds to the principle maxima.

$$\therefore \alpha = \pm \frac{(2n+1)\pi}{2}; n=1, 2, 3 \dots$$

$\therefore$ In general, $\boxed{e \sin \theta_n = \pm (2n+1) \frac{\lambda}{2}}$

is the condition for secondary maxima.



Intensity distribution: $I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2$

(1) Principle maxima ($\alpha = 0$)

$$I = I_0$$

(2) First secondary maxima ($\alpha = \pm \frac{3\pi}{2}$)

$$I = I_0 \left[\frac{\sin 3\pi/2}{3\pi/2} \right]^2 = \frac{I_0}{22} = \frac{4}{9\pi^2} = 0.045 I_0$$

$$\therefore \frac{I_1}{I_0} = \frac{1}{22} = 4.5\%$$

$$I_1 = 4.5\% \text{ of } I_0$$

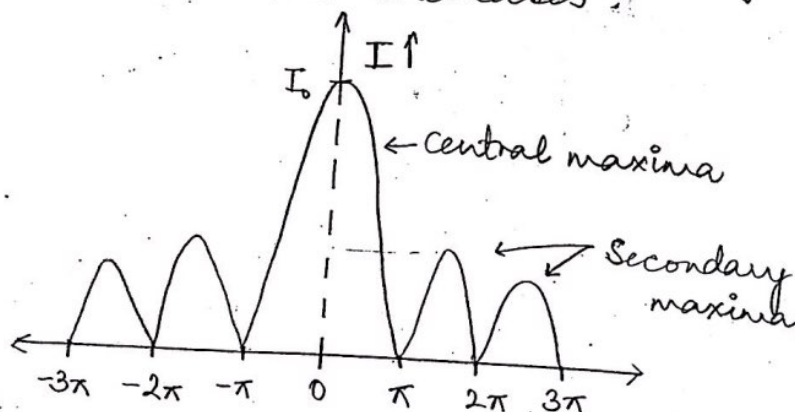
(3) Secondary maxima, $\alpha = \pm \frac{5\pi}{2}$

$$I_2 = I_0 \left[\frac{\sin \frac{5\pi}{2}}{\frac{5\pi}{2}} \right]^2 = \frac{4}{25\pi^2} I_0 = \frac{I_0}{62}$$

$$\therefore \frac{I_2}{I_0} = \frac{1}{62} = 1.61\%$$

$$\therefore I_2 = 1.61\% \text{ of } I_0$$

$\therefore$ The intensity of the principle maxima is the maximum and the intensity of the secondary maxima goes on decreasing very rapidly as the order increases.



$\therefore$ Intensities of the maxima $\propto \rightarrow$ decrease rapidly in the ratio,

$$1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2} : \frac{4}{49\pi^2} \dots$$

Resultant of n simple Harmonic motions:
 resolving R into parallel & perpendicular components:

$$R \cos \theta = a [1 + \cos d + \cos 2d + \dots + \cos (n-1)d] \quad \text{--- (1)}$$

$$R \sin \theta = a [\sin d + \sin 2d + \dots + \sin (n-1)d] \quad \text{--- (2)}$$

Multiplying eqn (1) by $2 \sin \frac{d}{2}$,

$$\begin{aligned} 2R \cos \theta \sin \frac{d}{2} &= a \left[2 \sin \frac{d}{2} + 2 \cos d \sin \frac{d}{2} + \dots + 2 \cos (n-1)d \sin \frac{d}{2} \right] \\ &= a \left[2 \sin \frac{d}{2} + \left\{ \sin \frac{3d}{2} - \sin \frac{d}{2} \right\} + \left\{ \sin \frac{5d}{2} - \sin \frac{3d}{2} \right\} + \dots \right. \\ &\quad \left. \dots \left\{ \sin \left(n - \frac{1}{2} \right) d - \sin \left(n - \frac{3}{2} \right) d \right\} \right] \end{aligned}$$

$$[\because 2 \cos A \sin B = \sin (A+B) - \sin (A-B)]$$

$$= a \left[\sin \frac{d}{2} + \sin \left(n - \frac{1}{2} \right) d \right]$$

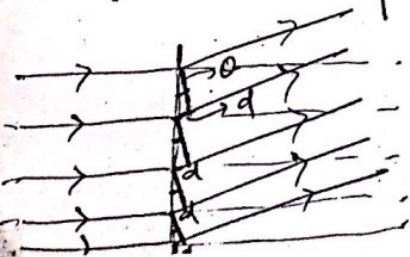
$$= 2a \sin \left(\frac{nd}{2} \right) \cos \left(\frac{(n-1)d}{2} \right) \quad \left[\because \sin a + \sin b = 2 \cos \left(\frac{a-b}{2} \right) \sin \left(\frac{a+b}{2} \right) \right]$$

$$\therefore R \cos \theta = a \frac{\sin \frac{nd}{2}}{\sin \frac{d}{2}} \cos \left(\frac{(n-1)d}{2} \right) \quad \text{--- (3)}$$

$$\text{Similarly, } R \sin \theta = a \frac{\sin \frac{nd}{2}}{\sin \frac{d}{2}} \sin \left(\frac{(n-1)d}{2} \right) \quad \text{--- (4)}$$

Sq & adding (3) & (4), $\Rightarrow R^2 = a^2 \frac{\sin^2 \frac{nd}{2}}{\sin^2 \frac{d}{2}}$

$$\text{or } R = \frac{a \sin \left(\frac{nd}{2} \right)}{\sin \left(\frac{d}{2} \right)}$$



∴ width of central maxima

let the distance of first secondary minima be x .

The width of central maxima is $2x$.

$$e \sin \theta = \pm n \lambda$$

(slit width = e)

$$e \sin \theta = \pm \lambda \quad (n=1)$$

$$\sin \theta = \pm \lambda / e$$

$$\theta = \sin^{-1}(\pm \lambda / e)$$

Hence central maximum extends between

$$\theta = \sin^{-1}(\lambda / e) \text{ to } \theta = \sin^{-1}(-\lambda / e)$$

∴ θ is the angular half width of the central maximum.

$$\sin \theta \approx \tan \theta = \frac{x}{D} \text{ for small } \theta.$$

$$\pm \frac{\lambda}{e} = \frac{x}{D}$$

$$x = \pm \frac{\lambda D}{e}$$

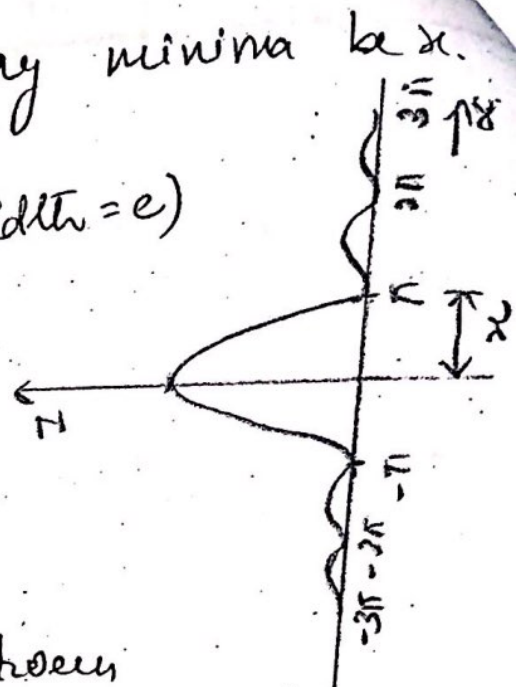
• If f is the focal length of the lens & the lens is very close to the slit, then $D = f$

$$x = \pm \frac{\lambda f}{e}$$

$$\therefore \text{width of central maxima} = x - (-x) = 2x$$

$$\boxed{\text{width of central maxima} = \frac{2 \lambda f}{e}}$$

⇒ width of central maximum is directly proportional to wavelength of light λ & inversely proportional to the slit width e .



Distribution curve

The intensity of Secondary maxima

$$I_1 = A^2 \frac{\sin^2 \frac{3\pi}{2}}{\left(\frac{3\pi}{2}\right)^2}$$

$$= \frac{4}{9\pi^2} A^2 = \frac{I_0}{22}$$

$$I_2 = A^2 \frac{\left(\sin \frac{5\pi}{2}\right)^2}{\left(5\pi/2\right)^2}$$

$$= \frac{4}{25\pi^2} A^2 = \frac{I_0}{61}$$

$$I_3 = \frac{4A^2}{49\pi^2} = \frac{I_0}{121}$$

The intensity of Secondary maxima falls off rapidly. Thus the relative intensities of the successive maxima are in the ratio!

$$1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2} : \frac{4}{49\pi^2} : \dots$$

or

$$1 : \frac{1}{22} : \frac{1}{61} : \frac{1}{121} : \dots$$

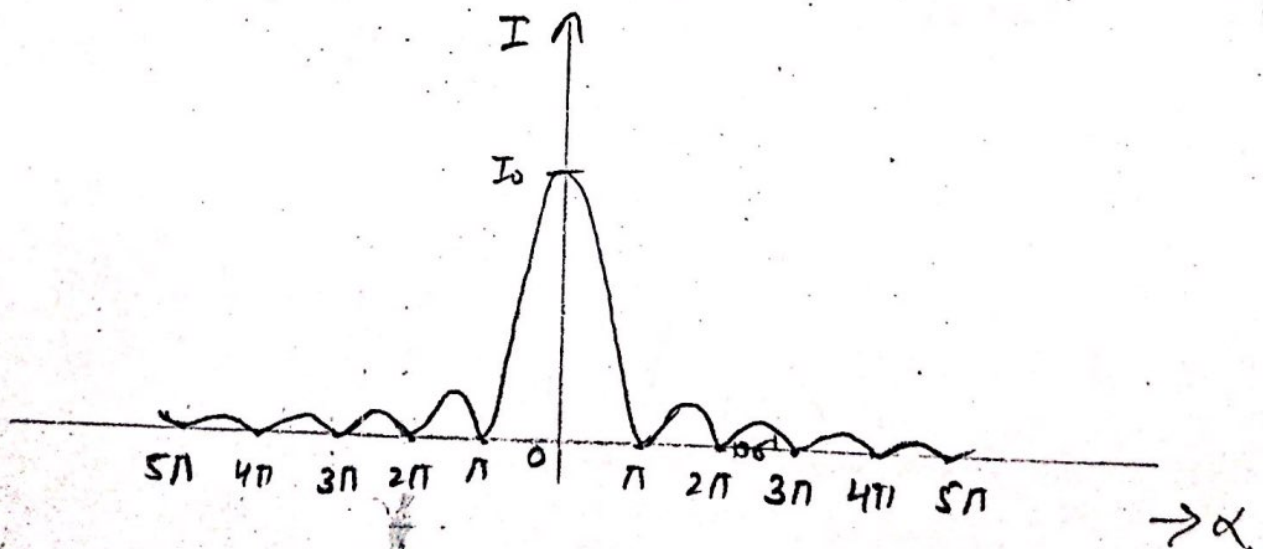
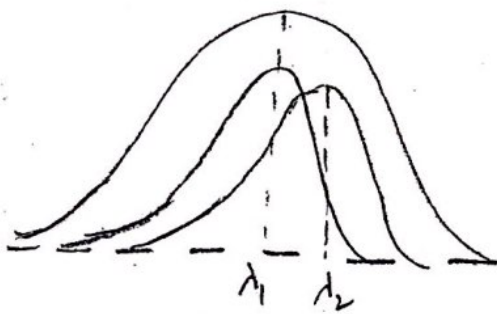


Fig. Plot of Intensities w.r.t values of α .

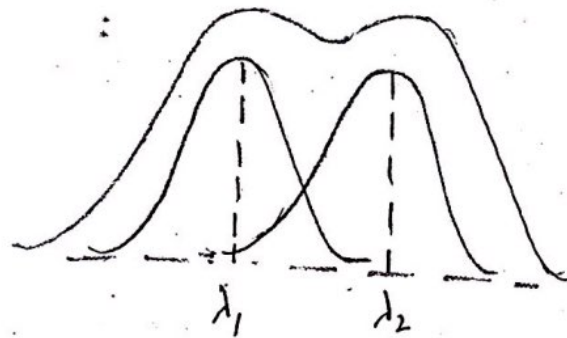
Give the Rayleigh criterion for resolution of two spectral lines & illustrate it with a diagram.
Resolution is defined as the separation between two very close objects.

The ability of an optical instrument to produce distinctly separate image of such close objects is called resolving power.

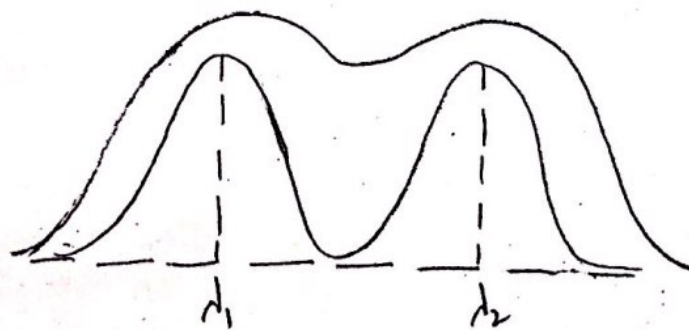
Rayleigh criteria:- Two closely spaced patterns may be regarded as separate or just resolved, if the central maxima of one coincided with the first minima of the other.



Unresolved



Just resolved



Completely
Resolved

What do you mean by absent spectra. What is the condition for absent spectra? (7)

For certain values of diffraction angle ' θ ' which satisfy the condition for principal maxima (i.e. $(e+f) \sin \theta = n\lambda$) absent from the grating spectrum

This happens for a given angle of θ the path difference between the 2 extreme rays is ' λ ' which satisfies the condition of minima for single slit. (i.e. $e \sin \theta = m\lambda$)

When the two conditions one for principal maxima for N-slit and another one for minima of single slit are satisfied simultaneously, a particular maxima of order n will be absent in the grating spectra. These are known as absent or missing spectra.

Condition for Absent Spectra is given by

$$\frac{e+f}{f} = \frac{n}{m}$$

* If $f=e$, then

$$n = 2m$$

$$n = 2, 4, 6, \dots \text{ for } m = 1, 2, 3, \dots$$

Hence, when width of transparencies and opacities of the grating are equal ($e=f$), then 2nd, 4th, 6th... order spectra are missing.

DIFFRACTION GRATING (FRAUNHOFER DIFFRACTION) AT N PARALLEL SLITS

What is a diffraction grating?

The arrangement of a large number of equidistant narrow rectangular slits of equal widths placed side by side parallel to one another is known as a diffraction grating.

- It consists of a well polished glass or metal surface on which a large number of fine equidistant parallel lines are ruled. The opaque portions of the grating are the roughened lines made by the ruling point & are called opacities as they do not allow light to pass through them. The clear spaces between the lines are known as transparencies which allow light to pass through them.

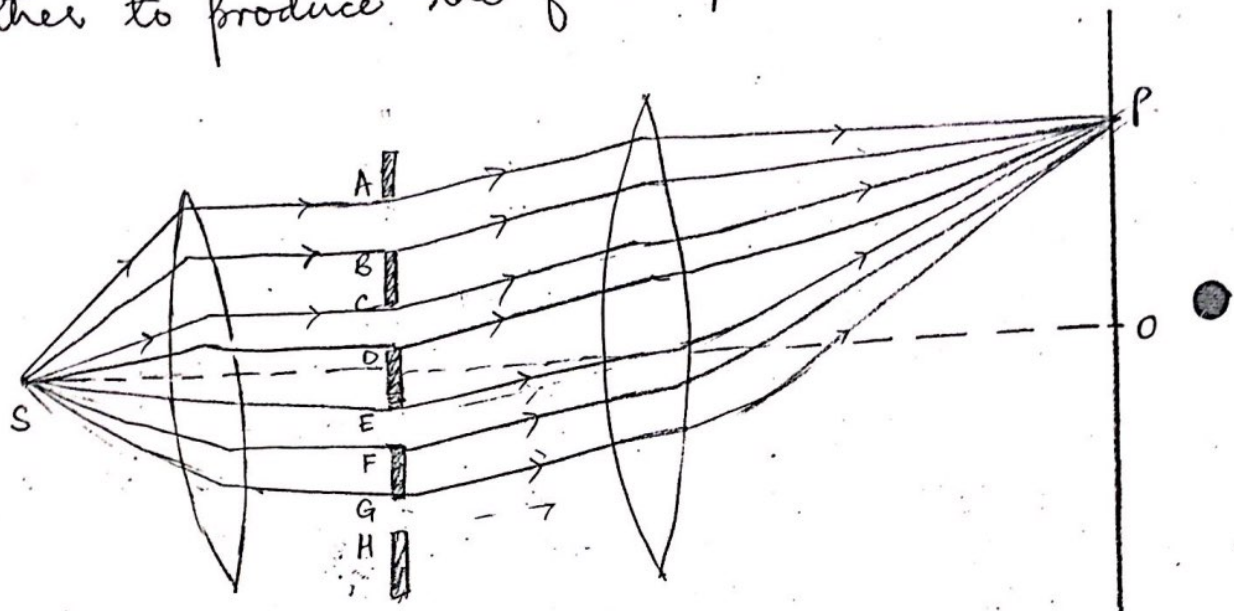
Grating element: The sum of the width of a transparency 'e' and the width of an opacity 'f' is known as the grating element of the grating.

Mathematically, grating element = $\frac{1}{\left(\text{No. of lines on the grating}\right)}$

$$\therefore (e+f) = \frac{1}{N}$$

RAYNHOFFER DIFFRACTION AT N-SLITS

Each of the N-slits send out a diffracted beam and these beams interfere with one another to produce the final pattern.



(e+f) : .. (grating element)

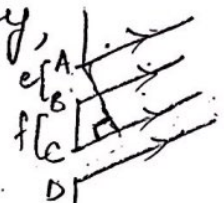
ABC...H represents a section of the plane transmission grating having N number of lines on it. A parallel beam of monochromatic light of wavelength λ is incident normally on it. The incident light is diffracted in various directions & are made to converge with the help of a convex lens on a screen placed in the focal plane of the lens.

All rays starting from AH (travelling along the direction of incident light) reach point O in phase & reinforce to produce the central maxima.

The rays diffracted at an angle θ with grating normal reach point P on passing through the lens in different phases.

If there are N slits, then the path difference between two consecutive slits is given by,

$$(e+f) \sin \theta$$



$\therefore$ Corresponding phase difference, $\delta = \frac{2\pi}{\lambda} (e+f) \sin \theta$.

$$\text{let } \frac{2\pi}{\lambda} (e+f) \sin \theta = 2\beta \text{ (say)}$$

$\therefore$ The resultant amplitude of ' n ' vibrations each having amplitude ' a ' ($= A \frac{\sin \alpha}{\alpha}$) with phase diff ' d ' from the vector addition of amplitude,

$$R = \frac{a \sin \frac{n d}{2}}{\sin \frac{d}{2}}$$

where, $a = \left(A \frac{\sin \alpha}{\alpha} \right)$ is the resultant amplitude from a single slit & $\alpha = \left(\frac{\pi e \sin \theta}{\lambda} \right)$

$$n = N \text{ \& } d = 2\beta$$

$$\therefore R = \left(A \frac{\sin \alpha}{\alpha} \right) \cdot \frac{\sin N\beta}{\sin \beta}$$

Corresponding intensity at point P,

$$I \propto R^2$$

$$\therefore I = \left(\frac{A \sin \alpha}{\alpha} \right)^2 \frac{\sin^2 N\beta}{\sin^2 \beta}$$

Term $\left(\frac{A \sin \alpha}{\alpha} \right)^2$ represents intensity distribution due to diffraction at a single slit.

Term $\left(\frac{\sin^2 N\beta}{\sin^2 \beta} \right)$ gives the distribution of

intensity in the diffraction pattern due to interference of waves from all N slits.

Principle Maxima:

$$R = \left(\frac{A \sin \alpha}{\alpha} \right) \frac{\sin N\beta}{\sin \beta}$$

Intensity maxima, $\sin \beta = 0 \Rightarrow \beta = \pm n\pi$
 $n = 0, 1, 2, \dots$

But if $\beta = \pm n\pi$ then $\sin N\beta = 0$

$\therefore \frac{\sin N\beta}{\sin \beta}$ becomes indeterminate.

$\therefore$ To avoid this,

$$\lim_{\beta \rightarrow \pm n\pi} \frac{\sin N\beta}{\sin \beta} = \lim_{\beta \rightarrow \pm n\pi} \frac{\frac{d}{d\beta} (\sin N\beta)}{\frac{d}{d\beta} (\sin \beta)}$$

$$\lim_{\beta \rightarrow \pm n\pi} \frac{N \cos N\beta}{\cos \beta} = \pm N$$

$$\therefore I_{\max} = \left(\frac{A \sin \alpha}{\alpha} \right)^2 N^2$$

$\therefore$ Intensity of principle maxima is directly proportional to N^2 & increases as the number of slits increases.

$$\text{As } \beta = \frac{\pi}{\lambda} (e+f) \sin \theta$$

$$\text{for } \beta = \pm n\pi,$$

$$\Rightarrow \frac{\pi}{\lambda} (e+f) \sin \theta = \pm n\pi$$

$$\text{or } \boxed{(e+f) \sin \theta = \pm n\lambda}$$

• For $n=0$, we get the zero order maximum

For $n=1, 2, 3, \dots$, we obtain first, second, third & so on principle maxima.

note: The $\pm$ sign shows that there are two principle maxima of the same order on either side of the zero order maxima.

$$\therefore N \cos N\beta \sin \beta - \sin N\beta \cos \beta = 0 \quad \text{or} \quad \sin N\beta = 0$$

But $\sin N\beta = 0$ corresponds to the positions of Minima.

$$\therefore \boxed{N \tan \beta - \tan N\beta = 0}$$

The roots of the above eqn other than those for which $\beta = n\pi$ (that corresponds to the principal maxima) give rise to another series of maxima known as the secondary maxima.

Intensity of secondary maxima:

$$\frac{\sin^2 N\beta}{\sin^2 \beta} = \frac{1}{\operatorname{Cosec}^2 N\beta} \cdot \frac{1}{\sin^2 \beta}$$

$$= \frac{1}{(1 + \cot^2 N\beta)} \cdot \frac{1}{\sin^2 \beta}$$

$$\left(\because \operatorname{Cosec}^2 \theta = 1 + \cot^2 \theta \right)$$

Multiplying & Dividing by N^2 ,

$$= \frac{N^2}{(N^2 + N^2 \cot^2 N\beta)} \cdot \frac{1}{\sin^2 \beta}$$

$$= \frac{N^2}{(N^2 + \cot^2 \beta)} \cdot \frac{1}{\sin^2 \beta} \quad \left(\because N \cot N\beta = \cot \beta \right)$$

$$= \frac{N^2}{N^2 \sin^2 \beta + \cos^2 \beta} \quad \left(\because \cot^2 \beta \times \sin^2 \beta = \cos^2 \beta \right)$$

$$= \frac{N^2}{N^2 \sin^2 \beta + (1 - \sin^2 \beta)} = \frac{N^2}{1 + (N^2 - 1) \sin^2 \beta}$$

Positions of Minima:

Intensity will be minimum when,

$$\sin N\beta = 0 \text{ but } \sin\beta \neq 0.$$

$$\Rightarrow \boxed{N\beta = \pm m\pi}$$

m can take values except $0, N, 2N, 3N, \dots$ because for these values, $\sin\beta = 0$ which gives position of principle maxima.

$$\therefore m = 1, 2, 3, \dots (N-1)$$

$$\therefore \frac{N\pi}{\lambda} (e + f) \sin\theta = \pm m\pi$$

$$\text{or } \boxed{N(e + f) \sin\theta = \pm m\lambda}$$

$\therefore$ There are $(N-1)$ minima positions between two principle maxima.

3) Secondary Maxima:

$$I = \frac{A^2 \sin^2 \alpha}{\lambda^2} \cdot \frac{\sin^2 N\beta}{\sin^2 \beta}$$

$$\text{Put } \frac{dI}{d\beta} = 0$$

$$\Rightarrow \frac{A^2 \sin^2 \alpha}{\lambda^2} \cdot 2 \frac{\sin^2 N\beta}{\sin \beta} \left[\frac{N \cos N\beta \sin \beta - \sin N\beta \cos \beta}{\sin^2 \beta} \right] = 0$$

$$\therefore \frac{\sin^2 N\beta}{\sin^2 \beta} = \frac{N^2}{1 + (N^2 - 1) \sin^2 \beta}$$

$\therefore$ Ratio of Intensity of secondary maxima to the intensity of the principal maxima {which varies as N^2 is,

$$\frac{N^2}{1 + (N^2 - 1) \sin^2 \beta} \bigg/ N^2 = \frac{1}{1 + (N^2 - 1) \sin^2 \beta}$$

$\Rightarrow$ ~~But~~ When N is large, the intensity of secondary maxima is weak and are invisible.

$\Rightarrow$ When N is small, secondary maxima may be observed.

For example: For $N=5$.

There will be 4 (ie, $(N-1)$) minima positions between each pair of principal maxima. Hence, there will be 3 (ie, $(N-2)$) positions of secondary maxima.

The resultant intensity at any point is the product of two factors $\left(\frac{\sin^2 \alpha}{\alpha^2}\right)$ and $\left(\frac{\sin^2 N\beta}{\sin^2 \beta}\right)$.

